

# PRD: "BitForge" Open-Source Protocol Emulator ASIC (Verilog + UVM Edition)

**Version:** 0.2 (draft) | **Date:** 2026-10-04 | **Target:** Tiny Tapeout IHP CMOS5L (130 nm), 6x4 tiles | **Deadline:** 2027-01-18 | **Shuttle:** March 2027

**Differences from v0.1:** the DUT is hand-written synthesizable **Verilog-2005** (no Hardcaml, no generators). Verification uses **SystemVerilog UVM 1.2 (IEEE 1800.2)** with SVA assertions. Requirement IDs are <AREA>-nnn . Priority: **M** Must, **S** Should, **C** Could.

## 1. Overview

### 1.1 Problem

Bit-banging protocols from a general CPU is timing-fragile; fixed peripherals are inflexible. Hardware debuggers and reverse engineers need a small, deterministic chip that can speak, sniff, and fuzz arbitrary wire protocols, including ones defined after fabrication.

### 1.2 Vision

A cycle-deterministic, multi-lane programmable I/O processor. Each lane is a minimal engine whose ISA is built around pin I/O, cycle counting, and bit shifting. Protocols are firmware, not silicon.

### 1.3 Goals

- 1. UART, SPI and I2C implemented purely in firmware.
- 2. New protocols possible post-fabrication within documented timing/I/O limits.
- 3. Fit 24 tiles (~24K cells) and close timing after full place and route.
- 4. Readable, reviewable Verilog RTL that is verified by a reusable UVM environment.
- 5. Fully open source with a reproducible flow.

### 1.4 Non-goals

Analog PHYs (USB/Ethernet transceivers, magnetics), protocols above ~100 Mbit/s, general-purpose compute.

### 1.5 Personas

| ID | Persona               | Need                                        |
|----|-----------------------|---------------------------------------------|
| P1 | Hardware debugger     | Sniff and decode buses, trigger on patterns |
| P2 | Reverse engineer      | Talk to or fuzz unknown serial interfaces   |
| P3 | Firmware author       | Write, simulate and load protocol firmware  |
| P4 | Verification engineer | Reuse UVM agents, sequences and coverage    |

## 2. Platform Constraints

| ID      | Constraint                                                                                                                                                                                     | Pri |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| CON-001 | Process: IHP 130 nm CMOS5L through Tiny Tapeout, using the <code>cmos5l</code> Verilog template                                                                                                | M   |
| CON-002 | <code>info.yaml</code> tile size 6x4; design must also build unchanged if 8x4 becomes available                                                                                                | M   |
| CON-003 | Interface limited to the Tiny Tapeout wrapper: <code>ui_in[7:0]</code> , <code>uo_out[7:0]</code> , <code>uio_in/out/oe[7:0]</code> , <code>clk</code> , <code>rst_n</code> , <code>ena</code> | M   |
| CON-004 | Budget ~1K cells/tile; leave headroom for clock tree and routing                                                                                                                               | M   |
| CON-005 | All design and verification source open source (Apache-2.0 RTL; verification code Apache-2.0)                                                                                                  | M   |
| CON-006 | Submission by 2027-01-18                                                                                                                                                                       | M   |
| CON-007 | DUT RTL is plain Verilog-2005 only, synthesizable by Yosys/OpenROAD flow; SystemVerilog is permitted <b>only</b> in the testbench, bind files and assertion modules, never in synthesized RTL  | M   |

## 3. Architecture Requirements

Reference: **N=4 lanes** sharing instruction memory, a pin crossbar, FIFOs and a host-control block.

| ID      | Requirement                                                                                                                                                                       | Pri |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| ARC-001 | 4 independent lanes (Verilog parameter <code>NUM_LANES</code> , 2-8), each with PC, registers, shifter and cycle counter                                                          | M   |
| ARC-002 | Shared instruction memory, at least 256 x 16-bit (target 512), with per-lane base and wrap addresses                                                                              | M   |
| ARC-003 | Instruction memory loadable at runtime via the host interface; each lane can be stopped, reprogrammed and restarted independently                                                 | M   |
| ARC-004 | Single clock domain; no gated or derived clocks                                                                                                                                   | M   |
| ARC-005 | Global "sync start" launches selected lanes on the same clock edge                                                                                                                | M   |
| ARC-006 | Instruction memory wrapped in one module ( <code>bf_imem</code> ) with interchangeable implementations (flip-flop, latch, SRAM macro) selected by <code>`define</code> /parameter | S   |
| ARC-007 | Per-lane TX and RX FIFOs, minimum 4 x 32-bit each, with level/empty/full flags                                                                                                    | M   |
| ARC-008 | 32-bit free-running timestamp counter, readable by lanes and host, latched                                                                                                        | S   |

|         |                                                                                                                                                                                                                                                                                                              |   |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|         | on pin-edge events                                                                                                                                                                                                                                                                                           |   |
| ARC-009 | 8 shared event flags usable for lane-to-lane coordination (set, clear, wait)                                                                                                                                                                                                                                 | M |
| ARC-010 | Reset puts all lanes in a halted state with defined register values; GPIOs tristate                                                                                                                                                                                                                          | M |
| ARC-011 | RTL partitioned into small documented modules with a module-level spec each: <code>bf_lane</code> , <code>bf_alu</code> , <code>bf_shift</code> , <code>bf_fifo</code> , <code>bf_xbar</code> , <code>bf_imem</code> , <code>bf_hif</code> , <code>bf_regs</code> , <code>bf_ts</code> , <code>bf_top</code> | M |

## 4. Instruction Set Architecture

Design principle: every instruction is **single-cycle and deterministic**, so timing equals instruction count.

| ID      | Requirement                                                                                                                                                       | Pri |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| ISA-001 | Fixed 16-bit instructions                                                                                                                                         | M   |
| ISA-002 | Non-stalling instructions execute in exactly 1 cycle; stall conditions documented cycle by cycle                                                                  | M   |
| ISA-003 | Per-lane registers: 4 x 16-bit general (X, Y, Z, W), 32-bit input shift register (ISR), 32-bit output shift register (OSR)                                        | M   |
| ISA-004 | Pin instructions: <code>IN</code> , <code>OUT</code> , <code>SET</code> , <code>DIR</code>                                                                        | M   |
| ISA-005 | <b>Side-set:</b> 1-3 configurable bits per instruction update pins in the same cycle as the main operation                                                        | M   |
| ISA-006 | <code>WAIT</code> on pin level/edge, event flag or timestamp, with optional timeout                                                                               | M   |
| ISA-007 | <code>JMP</code> conditions: <code>always</code> , <code>reg==0</code> , <code>reg!=0</code> (post-decrement), pin state, shift-count threshold, FIFO state, flag | M   |
| ISA-008 | 3-bit <code>DELAY</code> field (0-7 cycles) on every instruction                                                                                                  | M   |
| ISA-009 | <code>SHIFT</code> with direction, count 1-32, auto-push/auto-pull at configurable threshold                                                                      | M   |
| ISA-010 | <code>MOV</code> , <code>ADD/SUB</code> (immediate), <code>AND/OR/XOR/NOT</code> , bit-reverse, compare                                                           | M   |
| ISA-011 | <code>CALL</code> / <code>RET</code> with a 2-deep return stack                                                                                                   | S   |
| ISA-012 | Optional majority-vote oversampling on inputs (3-of-5)                                                                                                            | S   |
| ISA-013 | Host IRQ instruction raising a status bit and optional pin                                                                                                        | M   |
| ISA-014 | NRZI and bit-stuffing mode flags (for USB LS, CAN)                                                                                                                | C   |
| ISA-015 | Manchester encode/decode helper (for 10BASE-T)                                                                                                                    | C   |
| ISA-016 | Optional CRC unit (polynomial and width configurable, up to 16 bits plus                                                                                          | C   |

|         |                                                                                                                                                    |   |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------|---|
| ISA-016 | Optional CRC unit (polynomial and width configurable, up to 16 bits plus CRC-32 mode)                                                              | S |
| ISA-017 | Illegal opcode halts the lane and sets a sticky fault flag                                                                                         | M |
| ISA-018 | ISA specified in a single document and a machine-readable table ( <code>isa.yaml</code> ) used by the assembler, ISS and UVM instruction-generator | M |

## 5. I/O Requirements

| ID     | Requirement                                                                                                                                          | Pri |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| IO-001 | Up to 16 lane-controllable pins ( <code>uio</code> , <code>uo_out</code> ); <code>ui_in</code> sample-only except pins reserved for the host channel | M   |
| IO-002 | Crossbar register assigns each pin's output to a lane; any lane can read any pin                                                                     | M   |
| IO-003 | Multiple lanes on one pin: lowest lane index wins, sticky conflict flag set                                                                          | M   |
| IO-004 | Open-drain emulation per pin (output enable toggled, data held low), for I2C                                                                         | M   |
| IO-005 | 2-flop input synchronizer per pin with bypass mode; latency in cycles documented for both                                                            | M   |
| IO-006 | Optional input glitch filter (1-4 cycles) per pin                                                                                                    | S   |
| IO-007 | Per-pin input and output inversion                                                                                                                   | S   |
| IO-008 | All pins tristate in reset                                                                                                                           | M   |

## 6. Timing and Performance

| ID      | Requirement                                                                      | Pri |
|---------|----------------------------------------------------------------------------------|-----|
| TIM-001 | System clock $\geq 50$ MHz post-route (stretch 100 MHz); achieved fmax published | M   |
| TIM-002 | One pin transition per lane per clock ( $\geq 25$ MHz square wave at 50 MHz)     | M   |
| TIM-003 | Pin-edge to pin-response latency $\leq 4$ clocks in bypass mode                  | M   |
| TIM-004 | Per-lane 8.8 fractional clock divider; 115200 baud within 1% from 50 MHz         | M   |
| TIM-005 | Zero cycle skew between lanes started by sync start                              | M   |
| TIM-006 | Published table of maximum bit rate per protocol                                 | M   |

## 7. Host Interface

| ID      | Requirement                                                                                                                       | Pri |
|---------|-----------------------------------------------------------------------------------------------------------------------------------|-----|
| HIF-001 | SPI slave (SCK, CS, MOSI, MISO, 4 pins) for control                                                                               | M   |
| HIF-002 | Memory-mapped register space: program memory, lane config, FIFOs, flags, crossbar, status, timestamp                              | M   |
| HIF-003 | Burst write for program memory and burst read/write for FIFOs in one transaction                                                  | M   |
| HIF-004 | Interrupt/attention output pin                                                                                                    | M   |
| HIF-005 | Strap-selectable UART boot path                                                                                                   | C   |
| HIF-006 | Chip ID and version registers                                                                                                     | M   |
| HIF-007 | Debug visibility: per-lane PC, registers, halt, single-step                                                                       | S   |
| HIF-008 | PC breakpoint per lane                                                                                                            | C   |
| HIF-009 | Host SPI clock up to clk/4                                                                                                        | M   |
| HIF-010 | Register map defined once in a single spec file ( <code>regmap.yaml</code> ) and used to generate the UVM RAL model (see UVM-011) | M   |

## 8. Protocol Support

| ID      | Protocol                                                                     | Role                   | Pri |
|---------|------------------------------------------------------------------------------|------------------------|-----|
| PRO-001 | UART, 5-8 bits, parity, 1-2 stop, 300 baud to $\geq 3$ Mbaud                 | TX, RX, sniff          | M   |
| PRO-002 | SPI modes 0-3, up to clk/4, word size 1-32                                   | Master, slave, sniff   | M   |
| PRO-003 | I2C 100k/400k/1M, clock stretching                                           | Master, slave, sniff   | M   |
| PRO-004 | JTAG (TAP navigation, shift IR/DR)                                           | Master                 | S   |
| PRO-005 | SWD (request/ack/data with turnaround)                                       | Master                 | S   |
| PRO-006 | PS/2                                                                         | Host, device           | S   |
| PRO-007 | CAN 2.0 at $\leq 500$ kbit/s                                                 | Sniff, then TX         | C   |
| PRO-008 | USB low-speed 1.5 Mbit/s (NRZI, stuffing, CRC-16), external analog front end | Device-level handshake | C   |
| PRO-009 | 10BASE-T Ethernet (Manchester, link pulse, TX) via external PHY              | TX / loopback          | C   |
| PRO-010 | 1-Wire, WS2812, DShot, DMX, I2S                                              | Master                 | C   |
| PRO-011 | Multi-lane composition demo (clock lane + data lane)                         |                        | S   |

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## 9. Differentiating Capabilities

| ID      | Feature                                                                       | Pri |
|---------|-------------------------------------------------------------------------------|-----|
| DIF-001 | Triggered, timestamped, run-length-encoded edge capture (logic analyzer mode) | S   |
| DIF-002 | Auto-baud measurement                                                         | S   |
| DIF-003 | LFSR source for glitch/fuzz generation with deterministic seed                | S   |
| DIF-004 | Man-in-the-middle bridge mode using two lanes                                 | C   |
| DIF-005 | Pattern-match trigger output pin                                              | S   |
| DIF-006 | Reference firmware library, versioned                                         | M   |
| DIF-007 | Cycle-accurate ISS used as the UVM reference model, bit-for-bit with RTL      | M   |

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## 10. Software Tooling

| ID     | Requirement                                                                                                  | Pri |
|--------|--------------------------------------------------------------------------------------------------------------|-----|
| SW-001 | Assembler with labels, macros, side-set syntax and per-line cycle annotation                                 | M   |
| SW-002 | Disassembler                                                                                                 | S   |
| SW-003 | Cycle-accurate ISS with VCD pin waveform output; also built as a <b>DPI-C</b> library for the UVM scoreboard | M   |
| SW-004 | Python host library for loading, config and FIFO access via an RP2040 or FT232H bridge                       | M   |
| SW-005 | Firmware static timing analyzer                                                                              | C   |
| SW-006 | Example firmware per PRO item                                                                                | M   |

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## 11. RTL Design Requirements (Verilog)

| ID      | Requirement                                                                                                                                                                                                                                  | Pri |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| RTL-001 | Verilog-2005, synthesizable subset; one module per file, file name equals module name                                                                                                                                                        | M   |
| RTL-002 | Coding style guide published: synchronous active-low reset ( <code>rst_n</code> ), non-blocking assignments in sequential logic, no inferred latches in combinational logic, explicit default in case, no <code>initial</code> blocks in RTL | M   |
| RTL-003 | ...                                                                                                                                                                                                                                          | ... |

|         |                                                                                                                     |   |
|---------|---------------------------------------------------------------------------------------------------------------------|---|
| RTL-003 | All parameters and localparam constants (opcodes, register offsets) in a single header <code>bf_defs.vh</code>      | M |
| RTL-004 | Single clock domain; all asynchronous inputs through synchronizers (IO-005)                                         | M |
| RTL-005 | Zero unwaived Verilator <code>--lint-only -Wall</code> and Yosys warnings                                           | M |
| RTL-006 | Build profiles via defines: <code>BF_PROFILE_MIN</code> , <code>BF_PROFILE_STD</code> , <code>BF_PROFILE_MAX</code> | M |
| RTL-007 | Assertions and coverage live in separate bind ed SystemVerilog modules, so synthesized RTL stays pure Verilog       | M |
| RTL-008 | Every module has a header comment: purpose, ports, timing/latency, reset behavior                                   | S |
| RTL-009 | FPGA prototype (any open-toolchain board) runs PRO-001..003 against real devices                                    | S |

## 12. Verification Requirements (UVM)

### 12.1 Strategy and environment

| ID      | Requirement                                                                                                                                                                               | Pri |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| UVM-001 | UVM 1.2 testbench in SystemVerilog, top-level <code>tb_top</code> instantiating the DUT through the Tiny Tapeout wrapper ports                                                            | M   |
| UVM-002 | A reusable <code>bf_env</code> composed of agents, scoreboard, reference model, coverage collector and virtual sequencer                                                                  | M   |
| UVM-003 | Interfaces: SV <code>interface</code> for host SPI, one for the 24-pin GPIO bus, with modports and clocking blocks; DUT attached through virtual interfaces in <code>uvm_config_db</code> | M   |
| UVM-004 | Testbench is clock/reset aware with a dedicated <code>reset_agent</code> or reset sequence supporting mid-test reset                                                                      | M   |
| UVM-005 | All components factory-registered and configurable by <code>uvm_object config</code> classes; no hard-coded constants in components                                                       | M   |
| UVM-006 | Log/verbosity discipline: <code>uvm_info/warning/error/fatal</code> with defined ID naming; a test fails on any <code>UVM_ERROR</code> / <code>UVM_FATAL</code>                           | M   |

### 12.2 Agents

| ID      | Agent                                | Behavior                                                                                              | Pri |
|---------|--------------------------------------|-------------------------------------------------------------------------------------------------------|-----|
| UVM-007 | <code>spi_host_agent</code> (active) | Drives HIF-001 transactions; item with opcode, address, burst data; monitor reconstructs transactions | M   |

|         |                                                                                                                     |                                                                                                                                                     |   |
|---------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---|
| UVM-008 | pin_agent (active/passive per pin group)                                                                            | Drives ui_in / uio_in with programmable timing, glitch and jitter injection; monitor records edge-timestamped pin activity from uo_out / uio_out/oe | M |
| UVM-009 | uart_agent                                                                                                          | Protocol-compliant partner (TX/RX, configurable baud/parity/stop, error injection: framing, parity, break)                                          | M |
| UVM-010 | spi_dev_agent                                                                                                       | SPI master or slave partner, modes 0-3, configurable word size                                                                                      | M |
| UVM-011 | i2c_agent                                                                                                           | I2C master/slave partner with clock stretching, ACK/NACK, arbitration loss injection                                                                | M |
| UVM-012 | jtag_agent , swd_agent , ps2_agent                                                                                  | Partner models for the S-priority protocols                                                                                                         | S |
| UVM-013 | can_agent , usb_ls_agent , eth10_agent                                                                              | Partner models for stretch protocols                                                                                                                | C |
| UVM-014 | Every protocol agent includes a protocol-checker (monitor-side) that flags spec violations independently of the DUT | M                                                                                                                                                   |   |

### 12.3 Register model, reference model and scoreboards

| ID      | Requirement                                                                                                                                                                                   | Pri |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| UVM-015 | <b>RAL</b> ( uvm_reg_block ) generated from regmap.yaml , with adapter and predictor for the SPI host agent; built-in register tests (reset values, bit-bash, access) run                     | M   |
| UVM-016 | Reference model: ISS (SW-003) bound via <b>DPI-C</b> , stepping in lockstep with the DUT using the DUT's start/stop/sync events                                                               | M   |
| UVM-017 | lane_scoreboard : compares per-cycle pin output, PC trace, and FIFO traffic against the reference model; mismatches report cycle and lane                                                     | M   |
| UVM-018 | protocol_scoreboard : compares bytes/words observed by partner agents against the firmware's expected data, per protocol                                                                      | M   |
| UVM-019 | fifo_scoreboard : ordering, level and flag correctness for all FIFOs                                                                                                                          | M   |
| UVM-020 | Timing checker: measured edge-to-edge intervals are compared against ISA-defined cycle counts with zero tolerance for ISA-level timing and a configurable tolerance for protocol-level timing | M   |

### 12.4 Sequences and tests

| ID      | Requirement                                                                                                                                                                                                               | Pri |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| UVM-021 | Layered sequences: base item sequences, host-config sequences (load program, set crossbar, start lanes), protocol scenario sequences, virtual sequences coordinating host and pins                                        | M   |
| UVM-022 | <b>Constrained-random instruction generator:</b> legal instruction streams from <code>isa.yaml</code> , constraints for loop bounds, valid jumps, and weighted opcode mix, with directed bias options to hit corner cases | M   |
| UVM-023 | Directed tests: every opcode, every JMP condition, delay field extremes, side-set combos, auto-push/pull thresholds, stalls                                                                                               | M   |
| UVM-024 | Protocol tests: PRO-001..003 in all roles, sweeping baud/mode/speed, error injection, back-to-back and long-run                                                                                                           | M   |
| UVM-025 | Host-interface stress: random burst lengths, back-to-back transactions, CS glitches, illegal addresses                                                                                                                    | M   |
| UVM-026 | Multi-lane tests: sync start, flag handshakes, pin conflicts, simultaneous FIFO access                                                                                                                                    | M   |
| UVM-027 | Reset tests: reset during run, during burst, during FIFO traffic; state must return to known values                                                                                                                       | M   |
| UVM-028 | Illegal-opcode and fault-handling tests (ISA-017)                                                                                                                                                                         | M   |
| UVM-029 | Regression suite with named test lists: <code>smoke</code> (< 5 min), <code>nightly</code> , <code>full</code> ; each test seed logged and replayable                                                                     | M   |
| UVM-030 | Tests for S/C protocols (JTAG, SWD, PS/2, CAN, USB LS, Ethernet)                                                                                                                                                          | S   |

## 12.5 Coverage, assertions, formal

| ID      | Requirement                                                                                                                                                                                                                       | Pri |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| UVM-031 | Functional coverage model ( <code>covergroup</code> ) covering opcode x operand classes, JMP conditions x outcomes, shifter states, FIFO levels x events, pin events, crossbar configs, lane-pair interactions, register accesses | M   |
| UVM-032 | Cross coverage of protocol parameters (baud/mode/speed x error type x role)                                                                                                                                                       | M   |
| UVM-033 | Functional coverage goal $\geq 95\%$ , code coverage (line, branch, toggle, FSM) $\geq 90\%$ , each exclusion reviewed and documented                                                                                             | M   |
| UVM-034 | SVA assertion modules bound to the DUT: FIFO overflow/underflow, FSM legal state, reset values, tristate in reset, pin-conflict flag, stall cycle counts, HIF handshake rules                                                     | M   |
| UVM-035 | Assertions double as coverage ( <code>cover property</code> ) and are reported in the regression dashboard                                                                                                                        | S   |

|         |                                                                                                                         |   |
|---------|-------------------------------------------------------------------------------------------------------------------------|---|
| UVM-036 | Formal property checking on the SVA properties with SymbiYosys (or other available tool) for FIFO, HIF FSM and lane FSM | S |
| UVM-037 | Formal sequential equivalence of the pre- and post-synthesis netlist where tools permit                                 | C |

## 12.6 Gate-level, tools and process

| ID      | Requirement                                                                                                                                                                                                                                                                                                  | Pri |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| UVM-038 | The same UVM environment runs unchanged against the post-synthesis netlist and post-route netlist (with SDF back-annotation if available); reduced test list <code>gls_smoke</code> required to pass                                                                                                         | M   |
| UVM-039 | Simulator support matrix documented. Primary: a UVM-capable simulator available to the team (Questa, VCS, Xcelium, or Riviera-PRO). Secondary: Verilator with UVM support where available. RTL-only tests must also run with Icarus/Verilator using a non-UVM wrapper (cocotb or plain SV) for community use | M   |
| UVM-040 | CI runs lint, <code>smoke</code> regression, and coverage merge on every commit; <code>nightly</code> on schedule                                                                                                                                                                                            | M   |
| UVM-041 | Coverage database merged across seeds; a closure report is generated and archived with each release                                                                                                                                                                                                          | M   |
| UVM-042 | AI-assisted verification: LLMs may draft sequences, assertions, covergroups and bug triage; every artifact is human-reviewed and tagged <code>// AI-assisted</code> in the source with the prompt summary in the repo                                                                                        | S   |
| UVM-043 | Bug tracking: every RTL bug found by verification gets an issue with a reproducing test and seed, and a regression test added                                                                                                                                                                                | M   |
| UVM-044 | Post-silicon bring-up tests: self-test firmware, loopback checks, and expected waveforms derived from UVM tests                                                                                                                                                                                              | M   |
| UVM-045 | Verification plan (vPlan) document mapping each requirement ID in this PRD to tests, assertions and coverage items, kept up to date and checked at release                                                                                                                                                   | M   |
| UVM-046 | Mutation sampling: inject at least 20 RTL mutants (inverted conditions, off-by-one cycle counts) and report the kill rate                                                                                                                                                                                    | C   |

## 13. Physical Design

| ID      | Requirement                                                                    | Pri |
|---------|--------------------------------------------------------------------------------|-----|
| PHY-001 | Post-synthesis area $\leq$ 65% of nominal 6x4 area at <code>STD</code> profile | M   |
| PHY-002 | Full Tiny Tapeout flow completes without DRC/LVS errors                        | M   |

|         |                                                             |   |
|---------|-------------------------------------------------------------|---|
| PHY-003 | Setup and hold met at target clock across supported corners | M |
| PHY-004 | Synthesis area/fmax tracked in CI from week 2               | M |
| PHY-005 | Power estimate documented                                   | S |
| PHY-006 | MIN profile fallback that fits comfortably                  | M |

## 14. Documentation and Open Source

| ID      | Requirement                                                                             | Pri |
|---------|-----------------------------------------------------------------------------------------|-----|
| OSS-001 | Public repo with permissive license from day one                                        | M   |
| OSS-002 | Datasheet: pinout, registers, ISA, timing, limits                                       | M   |
| OSS-003 | Tutorials: "UART in 10 instructions", "add a new protocol"                              | S   |
| OSS-004 | One-command build to GDS; one-command regression launcher ( make smoke , make nightly ) | M   |
| OSS-005 | Dev-board and analog front-end notes                                                    | S   |
| OSS-006 | Verification user guide: how to write a new agent/sequence for a new protocol           | S   |
| OSS-007 | Contribution guide and issue templates                                                  | S   |

## 15. Repository Structure (informative)

rtl/ (pure Verilog) | rtl/bf\_defs.vh | veril/uvm/{agents,env,seq,tests,ral,cov} |  
verif/sva/ | veril/formal/ | model/iss/ (C + DPI) | sw/{asm,host,fw} | spec/  
{isa.yaml,regmap.yaml,vplan} | flow/ (Tiny Tapeout info.yaml,constraints) | docs/

## 16. Milestones

| Milestone | Date       | Exit criteria                                                                                                       |
|-----------|------------|---------------------------------------------------------------------------------------------------------------------|
| M0        | 2026-10-18 | ISA and regmap v0.1 frozen; repo, CI, flow template running; vPlan skeleton                                         |
| M1        | 2026-11-01 | Single-lane Verilog + assembler + ISS; UVM env with SPI host + pin agents; UART passes                              |
| M2        | 2026-11-22 | 4 lanes, crossbar, FIFOs, HIF; RAL tests pass; SPI and I2C agents/ tests; first area/timing report                  |
| M3        | 2026-12-13 | Scoreboard with DPI reference model; random instruction regression; coverage > 85%; assertions + formal; FPGAs demo |

|    |            |                                                                     |
|----|------------|---------------------------------------------------------------------|
|    |            | regression; coverage > 85%; assertions + formal; FPGA demo          |
| M4 | 2027-01-04 | Coverage closure, GLS pass, P&R closure, S-level protocols          |
| M5 | 2027-01-18 | Submission: docs, datasheet, final GDS, vPlan sign-off, release tag |

17. Risks

| ID      | Risk                                                                                                      | Mitigation                                                                                                       |
|---------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| RSK-001 | Instruction memory area overrun                                                                           | ARC-006 swappable memory; MIN profile; early synthesis                                                           |
| RSK-002 | Timing closure below 50 MHz                                                                               | Shallow single-cycle datapath; pipeline decode if needed                                                         |
| RSK-003 | Pin count limits protocols                                                                                | Flexible crossbar; document pin budgets                                                                          |
| RSK-004 | USB/Ethernet analog needs                                                                                 | Keep stretch; define external front end                                                                          |
| RSK-005 | <b>UVM requires a commercial or partially supported simulator</b> , which community contributors may lack | UVM-039: non-UVM smoke path on open tools; clear matrix; check availability of Riviera-PRO/Questa licenses by M0 |
| RSK-006 | DPI-C integration differs per simulator                                                                   | Isolate DPI behind a thin API; test on at least two simulators                                                   |
| RSK-007 | Verification scope exceeds schedule                                                                       | Prioritize M items; vPlan drives ordering; reuse agents across protocols                                         |
| RSK-008 | Hand-written Verilog has no generator to enforce consistency across ISA, assembler and docs               | ISA-018 and HIF-010: shared YAML specs plus a CI consistency check that decode tables in RTL match isa.yaml      |
| RSK-009 | Tile size changes (8x4)                                                                                   | CON-002; build profiles scale                                                                                    |

18. Success Criteria

1. **SC-1:** BPO-001, 002 pass in RTL, gate level and FPGA